

## Savitribai Phule Pune University

## T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination

## Lab Course CS-321-VSC-P

## Foundation of Artificial Intelligence and Machine Learning

Duration: 3 Hours

Maximum Marks:35

**Q1)** Write a program to implement Means-End Analysis for solving a goal-based problem **[10 marks]**

**Q2)** Write a program to implement the Random Forest Classifier for classification tasks. **[20 marks]**

```
dataset = { 'Study_Hours': [2, 3, 4, 5, 6, 7, 1, 2, 5, 8],
            'Attendance': [60, 65, 70, 75, 80, 85, 55, 62, 78, 90],
            'Assignment_Score': [55, 60, 65, 70, 75, 80, 50, 58, 72, 85],
            'Result': ['Fail', 'Fail', 'Pass', 'Pass', 'Pass',
                       'Pass', 'Fail', 'Fail', 'Pass', 'Pass'] }
```

OR

**Q2)** Write a program to demonstrate hyperplane classification using a Support Vector Machine on the iris dataset. Create a Pipeline containing a StandardScaler and a LinearSVC (with c =1 and use hinge loss). **[20 marks]**

| No. | Sepal Length | Sepal Width | Petal Length | Petal Width |
|-----|--------------|-------------|--------------|-------------|
| 1   | 5.1          | 3.5         | 1.4          | 0.2         |
| 2   | 4.9          | 3.0         | 1.4          | 0.2         |
| 3   | 4.7          | 3.2         | 1.3          | 0.2         |
| 4   | 4.6          | 3.1         | 1.5          | 0.2         |
| 5   | 5.0          | 3.6         | 1.4          | 0.2         |
| 6   | 7.0          | 3.2         | 4.7          | 1.4         |
| 7   | 6.4          | 3.2         | 4.5          | 1.5         |
| 8   | 6.9          | 3.1         | 4.9          | 1.5         |
| 9   | 5.5          | 2.3         | 4.0          | 1.3         |
| 10  | 6.5          | 2.8         | 4.6          | 1.5         |

**Q3)** Viva **[05 marks]**